STONE ČECH COMPACTIFICATION

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1. motivation

Given a top space X, we want the *best possible* approximation of X that is compact and hausdorff. We essentialy prove the existence of left adjoint functor to the inclusion functor from category of compact hausdorff topological spaces to the category of topological spaces.

2. partial order

The obvious way to construct a compactification[1] is by embedding in a compact space[2] and taking the closure of the image, but this need not necessitate that all continuous functions extend, for example $(0,1)$ and $f(x) = sin(1/x)$ is a continuous function from $(0,1)$ to $\mathbb{R}^2$ that cannot be extended to $[0,1]$. The set of compactifications is therefore partially ordered by the sets of continuous functions they extend. This can be equivalently stated as $K_1 \leq K_2$ iff there exists a surjective continuous map $q : K_2 \to K_1$.

3. construction

Once we have the poset, the existence of the stone čech compactification dependes on the existence of the *maximum* element. In language of order theory, the set of all compactifications of a tychonoff space X forms a complete upper semilattice. Let the set of all compactifications be $\mathcal{K} = \{(K_i, e_i)\}_{i\in I}$. Let $P = \prod_{i\in I} K_i$ [3] and $E(x) = (e_i(x))_{i\in I}$. E is an embedding and we define $K_{max} = \overline{E(X)}$. K_{max} dominates K_i for all $i \in I$ as we have $\Pi_i : P \to K_i$ and $\Pi_i|_{K_{max}} : K_{max} \to K_i$ and it is an easy exercise to show this map is surjective and fixes X.

4. looking at it via limit points

If $K_1 \leq K_2$ then $q : K_2 \to K_1$, we can show $q(K_2 \setminus X) = K_1 \setminus X$.[4] The obvious thing to try now is: to collapse a subset of $K \setminus X$ to a point and use the quotient topology on the new space to get a new compactification which is $\leq K$. The issue arises at hausdorfness, which in general does not behave well with quotient maps. We can collapse a subset of the boundary to a point iff it is closed in the overarching space [5]. An additional result is that a space is open in its compactification iff it is locally compact. Therefore a one point compactification exists iff the space is locally compact.

5. on tychonoff's cube

Moving up the partial order keeps boundary points separate. Any two boundary points can be separated via a continuous functionto $[0,1]$. We embed X into the product space:

$$P = \prod_{f\in J} [0,1]$$

where J is the set of all continuous functions from X to $[0,1]$ via the evaluation map $e(x) = (f(x))_{f\in J}$. Let K be a compactification of X. Because K is a compact hausdorff space it can be embedded into a tychonoff cube indexed by $J_K = C(K, [0,1])$. So, K sits faithfully inside $\prod_{g\in J_K}[0,1]$. Every continuous function g on K restricts

[1] whenever compactification is used it means the new space is compact and hausdorff.

[2] An embedding in a compact space exists only for tychonoff spaces. This forces the space X to be tychonoff.

[3] for the sake of thouroughty, we are using tychonoff's theorem here to say the product is compact.

[4] The converse is obviously false as you would need the extended function between compactifications to be continuous.

[5] We are using the following general result here: The quotient space X/A (formed by identifying all points in A to a single point $[A]$) is a Hausdorff space if and only if A is closed in X, and for every $x \notin A$, there exist disjoint open sets $U, V \subseteq X$ such that $x \in U$ and $A \subseteq V$.

to a continuous function f on X. Therefore, J_K is just a strict subset of J (all bounded functions on X), hence there is a natural, continuous projection map:

$$\pi : \prod_{f \in J} [0,1] \to \prod_{g \in J_K} [0,1]$$

When you restrict this projection π to the closure space K_{max}, it maps perfectly onto K.

$$q = \pi|_{K_{max}} : K_{max} \to K$$

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